Time Series

- Sometimes available data is of type time series ie. measurements are related to successive intervals of time. For eg. data related to yearly, quarterly or monthly figures.
- In time series analysis, past is analyzed in order to have a better understanding of the future. Moreover, comparisons of developments of two or more series over time can be performed.

- Utility of Time series analysis
  - It helps in understanding past behavior
  - It helps in planning future operations: Statistical techniques can be used to analyse time series.
  - It helps in evaluating current accomplishments: Actual performance can be compared with expected performance and cause of variation can be analysed.
  - It facilitates comparison.

  Time series analysis can not tell about future events with 100% accuracy.

Components of Time series

- Fluctuations of a time series can be classified into four basic types of variations. These fluctuations are called components of time series:
  - Secular trend
  - Seasonal variations
  - Cyclical variations
  - Irregular variations

- Secular Trend: Secular or long term trend is the basic tendency to grow or decline over a period of time.
  - Trends are divided under two heads:
    - Linear or straight line trend
    - Non-Linear trend
  - Longer the period covered, more significant the trend is. Roughly the period must cover two or three complete cycles.
  - It is not necessary that rise or fall must continue in the same direction throughout the period. General tendency of the data is to be observed.

- Seasonal variations: Periodic movements which occur regularly every year and have their origin in the nature of the year itself.
- These events can be predicted accurately.
- Amplitude of seasonal variations may vary but their period is fixed being one year.
- Factors that cause seasonal variations:
  - Climate and weather conditions: For eg. Rainfall, humidity etc. For eg. during winter there is demand for woollen clothes. In agriculture, there are two seasons, growing season and harvesting season that affect the income of the farmer.
  - Customs, traditions, habits: For eg. during diwali, dushhehra, there is big demand for sweets and cash also. On the first of every month, there are heavy withdrawals in banks
- An accurate knowledge of seasonal behavior can help in removing seasonal effects. For eg. By introducing diversified products having different seasonal peaks can increase profit.

- Cyclical variations: Recurrent variations in time series that usually last longer than a year and are regular neither in amplitude nor in length.
- Cyclical fluctuations are long term movements that represent consistently recurring rises and declines in activity.
- In business cycle there are four well defined period or phases: prosperity, decline, depression and improvement. These phases are followed in the order given.
- Study of cyclical variations is useful in avoiding fluctuations.
- These variations are the most difficult type of variations to measure because of two reasons:
  - Business cycles do not show regular periodicity.
  - Business cyclical variations are mixed with random or irregular forces which makes it impracticable to isolate the effect of cyclical variations.

- Irregular variations: Also called erratic, accidental or random variations.
- These are the variations that do not repeat in a definite pattern.
- It includes all types of variations other than trend, seasonal and cyclical.
- Chance factor is included in irregular variations. For eg. Floods, earthquakes, war etc.
- While measuring variations, trend and seasonal variations are measured separately and cyclical and irregular variations are measured separately.
- All the four variations are not independent of each other. For eg. Seasonal variations may aggrevate cyclical variations.

Primary adjustments before analysing time series

- Before analysing time series, make sure that figures are comparable over time.
- Common type of variations that make data non-comparable:
  - Calendar variation
  - Price changes
  - Population changes.
  - Miscellaneous changes: Units in which data are reported may have changed. There may be change in definition, classification of product.

Measurement of Trend

- Reasons for trend measurement:
  - To find out trend characteristics. For eg. Comparing the growth of textile industry with the growth in economy.
  - To enable us to eliminate trend in order to study other elements.
- Method of moving averages.

- Merits of the method:
  - Simple
  - Flexible as if few more figures are added, entire calculations are not changed.
  - If period of moving average happens to coincide with the period of cyclical fluctuations in the data, such fluctuations are automatically eliminated.
  - Shape of trend is determined by the data rather than statistician's choice of function.
  - Particularly effective if the trend of a series is very irregular.
- Limitations:
  - The longer the period of moving average, greater the number of years for which trend values cannot be obtained.
  - No hard and fast rule for the choice of period. One has to use his own judgement.
  - Moving average is not represented by a mathematical function, this method can not be used in forecasting.
  - Theoretically it is said that if period of moving average happens to coincide with the period of cycle, cyclical fluctuations are eliminated, but practically cycles are not perfectly periodic.
  - When the trend situations is not linear, moving average lies either above or below the true sweep of the data.

Forecasting

- When estimates of future conditions are made on a systematic basis, the process is called forecasting and the figure or statement obtained is known as a forecast.
- Forecasting aims at reducing the areas of uncertainty that surround management decision making with respect to costs, profit, sales, production, pricing, capital investment and so on.
- Future events are rarely assured so organized system of forecasting is necessary.

Role of forecasting in business:

- In many respects, future tends to move like past. Forecasting helps in planning future business activities.
- It is impossible to forecast the future precisely, there always must be some range of error allowed.

Steps in Forecasting

- Steps in forecasting:
  - Understanding why changes in the past have occurred: It necessary to get ‘speedometer reading’ of the current rate and how fast this rate is increasing or decreasing ie. Acceleration and Deceleration.
  - Determining which phases of business activity must be measured: Measure certain phases of business activity in order to predict what changes will probably follow.
  - Selecting and compiling data to be used as measuring devices: Statistical data cannot be collected and analysed in an intelligent manner unless there is a sufficient understanding of business fluctuations.
  - Analysing the data: Data are analysed in the light of one’s understanding of the reason why change occurs.

Methods of Forecasting

- Attempts are being made to make forecasting as scientific as possible. Base of scientific forecasting is statistics.
- Techniques of forecasting:
  - Business Barometers: Data series that can be used as index or indicator. Some of the important series:
    - Wholesale prices
    - Consumer credit.

- Extrapolation: In this technique the most reasonable expectation is that the variable will follow its already established path.
  - Problem is to determine accurately the appropriate trend curve and the values of its parameters. Some techniques are:
    - Arithmetic trend: Straight line arithmetic trend assumes that growth will be by a constant absolute amount each year.
    - Semi-log trend: It assumes a constant percentage increase each year. It translates into a straight line when drawn within a logarithmic vertical scale.
    - Modified exponential trend: Cruve assumes that each increment of growth will be a constant percent less that 100 of the previous one.
    - Logistic curve: It has both an upper limit and a lower limit. Gradual slow done of growth is there as maturity is approached.
    - Gompertz curve: Same as logistic curve but used to describe growth of industrial output.
  - That curve is selected that best fits the past movement of data.

- Regression Analysis: When two variables are functionally related then knowledge of one will make possible an estimate of the other.
  - Simple regression and multiple regression can be performed.
  - Two dangers in using regression analysis:
    - When there is spurious correlation among time series.
    - There is the risk that the estimated regression is false.
- Econometric Models: These models include application of mathematical economic theory and statistical procedures to economic data in order to verify economic theorems and to establish quantitative results.
  - An econometrician is basically an economist, a statistician and a mathematician.
  - These models take the form of simultaneous equations. Values of constants are supplied by a study of statistical time series.
  - Development of the model requires sufficient data so that correct relationships can be established. Whenever data are scarce, this method can not be profitably applied.
  - These models are not very popular as it is very difficult for forecaster to construct its own model of economy. Already developed models(research models) are used.

- Forecasting by the use of time series analysis:
  - Following variations can be measured:
    - Seasonal
    - Cyclical
    - Trends
    - Growth rates of trends.
  - Combined effect of trends, cycles and irregular is difficult to measure.
  - This method can be used only when several years data are available and when relationships and trends are both clear and relatively stable.
- Opinion Polling: Survey of opinion of experts.
  - For eg. Survey of opinion of sales representatives, wholesalers etc.
  - It can be very helpful in formulating demand projections.

- Causal Models:
  - Most sophisticated kind of forecasting tool.
  - Expresses mathematically the relevant causal relationships.
  - It may also directly incorporate the results of time series analysis.
  - This model takes into account everything known of the dynamics of the flow system and utilises predictions. This model generally includes factors for each location in the flow chart to describe overall product flow.
  - If certain type of data is lacking, assumptions can be made and then it is tracked whether assumptions were true or not.
  - This model is continually revised as more knowledge becomes available.

- Exponential Smoothing:
  - This is a special kind of weighted average and is found extremely useful in short term forecasting.
  - To forecast the value of a time series for the period t+1, a function is used that takes two components: the actual value of the series for period t and the forecasted value for the same period made in the period t-1
  - Smoothing process:
    - Exponentially smoothed value at time period t is denoted by Si.
      - » $S_2=aX_2+BS_1$ ie. $St=aX_t+BS_{(t-1)}$
    - a and B are smoothing constants and a+B =1. Rule of thumb for taking the value of a is when the magnitude of random variations in the series is large, small a should be selected, otherwise if moderately stable process is there then large a should be selected.
    - $\Delta S_t=S_t-S_{(t-1)}$
    - $T_t=a\Delta S_t+BT_{(t-1)}$
    - Forecast for t+1 is $S'_t=S_t+(B/a)T_t$

- Survey method:
  - Field surveys are conducted to obtain necessary information both quantitative and qualitative.
  - Important limitation is that information supplied may be biased. Bias can be reduced by proper designing the questionnaires, verifying results etc.
- Choice of method of forecasting:
  - It depends on many factors:
    - Context of the forecast
    - Relevance and availability of historical data
    - Degree of accuracy desired
    - Time period for which forecasts are required
    - Cost/Benefit of forecast
    - Time available for making the analysis.

Reference(s)

- Gupta, S. P., and M. P. Gupta. *Business statistics*. Sultan Chand & Sons, 1983.